

Úloha 1

Pokažte, že v libovolném kompl. v. pr. V se sk. součinem $\langle \cdot, \cdot \rangle$ platí $\forall u, v \in V$

$$\textcircled{1} \operatorname{Re} \langle u, v \rangle = \frac{1}{4} (\|u+v\|^2 - \|u-v\|^2)$$

$$\textcircled{2} \operatorname{Im} \langle u, v \rangle = \frac{1}{4} (\|u-iv\|^2 - \|u+iv\|^2)$$

$$\begin{aligned} \textcircled{1} \quad \|u+v\|^2 &= \langle u+v, u+v \rangle \stackrel{\text{SLZ}}{=} \langle u+v, u \rangle + \langle u+v, v \rangle = \\ &= \langle u, u \rangle + \langle v, u \rangle + \langle u, v \rangle + \langle v, v \rangle = \|u\|^2 + \|v\|^2 + \langle v, u \rangle + \langle u, v \rangle \\ \|u-v\|^2 &= \langle u-v, u-v \rangle = \langle u-v, u \rangle + \langle u-v, v \rangle = \\ &= \langle u, u \rangle + \langle -v, u \rangle + \langle u, v \rangle + \langle -v, v \rangle = \\ &= \|u\|^2 + (-1) \langle v, u \rangle + \langle u, v \rangle + (-1) \langle v, v \rangle = \\ &= \|u\|^2 - \langle v, u \rangle - \langle u, v \rangle + \|v\|^2 \\ \|u+v\|^2 - \|u-v\|^2 &= \|u\|^2 + \|v\|^2 + \langle v, u \rangle + \langle u, v \rangle - (\|u\|^2 - \|v\|^2 - \langle v, u \rangle - \langle u, v \rangle) = \\ &= 2\langle v, u \rangle + 2\langle u, v \rangle = 2(\overline{\langle u, v \rangle} + \langle u, v \rangle) \end{aligned}$$

~~Pro~~ $\forall x \in \mathbb{C}$ platí $\bar{x} + x = 2 \operatorname{Re}(x)$, protože $\begin{matrix} x = a+bi \\ \bar{x} = a-bi \end{matrix}$

$$a+bi + a-bi = 2a, \text{ kde } a = \operatorname{Re}(x)$$

$$\|u+v\|^2 - \|u-v\|^2 = 2 \cdot 2 \operatorname{Re} \langle u, v \rangle \Rightarrow \underline{\operatorname{Re} \langle u, v \rangle = \frac{1}{4} (\|u+v\|^2 - \|u-v\|^2)}$$

$$\begin{aligned}
 (2) \|u - iv\|^2 &= \langle u - iv, u - iv \rangle = \langle u - iv, u \rangle + \langle u - iv, -iv \rangle = \\
 &= \langle u, u \rangle + \langle -iv, u \rangle + \langle u, -iv \rangle + \langle -iv, -iv \rangle = \\
 &= \|u\|^2 + (-i)\langle v, u \rangle + (-i)\langle u, v \rangle + (-i) \cdot (-i)\langle v, v \rangle = \\
 &= \|u\|^2 + \|v\|^2 + i\langle v, u \rangle - i\langle u, v \rangle
 \end{aligned}$$

$$\begin{aligned}
 \|u + iv\|^2 &= \langle u + iv, u + iv \rangle = \langle u + iv, u \rangle + \langle u + iv, iv \rangle = \\
 &= \langle u, u \rangle + \langle iv, u \rangle + \langle u, iv \rangle + \langle iv, iv \rangle = \\
 &= \|u\|^2 + (i)\langle v, u \rangle + i\langle u, v \rangle + i \cdot i\langle v, v \rangle = \\
 &= \|u\|^2 + \|v\|^2 - i\langle v, u \rangle + i\langle u, v \rangle
 \end{aligned}$$

$$\|u - iv\|^2 - \|u + iv\|^2 = \cancel{\|u\|^2} + \cancel{\|v\|^2} + i\langle v, u \rangle - i\langle u, v \rangle - \cancel{\|u\|^2} - \cancel{\|v\|^2} + i\langle v, u \rangle - i\langle u, v \rangle$$

$$= 2i\langle v, u \rangle - 2i\langle u, v \rangle = 2i(\langle \bar{u}, v \rangle - \langle u, v \rangle) =$$

$$= 2(i(\langle \bar{u}, v \rangle - \langle u, v \rangle))$$

$$\forall x \in \mathbb{C}: i(\bar{x} - x) = 2\operatorname{Im}(x), \text{ donde } \begin{matrix} x = a + bi \\ \bar{x} = a - bi \end{matrix}$$

$$i(a - bi - a - bi) = i(-2bi) = -2b \cdot (i)^2 = -2 \cdot (-1) \cdot b = 2 \cdot b, \text{ kde } b = \operatorname{Im}(x)$$

$$\|u - iv\|^2 - \|u + iv\|^2 = 2 \cdot 2\operatorname{Im}(\langle u, v \rangle) \Rightarrow \operatorname{Im}(\langle u, v \rangle) = \frac{1}{4}(\|u - iv\|^2 - \|u + iv\|^2)$$

